Yung-Tse (Mes) Cheng 鄭詠澤

Taiwan

GitHub | LinkedIn | Website

Summary

Computer Science undergraduate focused on Linux graphics virtualization, C++ systems programming, and GPU software stacks. Implemented VirtIO-GPU and input support for semu, presented at Open Source Summit North America 2026 and COSCUP 2026, and managed infrastructure for the NCU Mathematics Department.

Education

National Taiwan Normal University Sep 2024–Present
B.S. in Computer Science and Information Engineering

- **Member**, Computer Graphics Laboratory
  - Presented VirtIO-GPU virtualization work at Open Source Summit North America 2026 and COSCUP 2026.

National Central University Sep 2020–Jun 2024
Undergraduate Studies in Mathematics (Transferred)

- **Systems and Network Manager**, Department of Mathematics
  - Rebuilt the department's server cluster by migrating services from Kubernetes to Proxmox VE and Docker.
  - Maintained web services and servers and led a student team delivering Higher Education Sprout projects.

Open Source Projects

semu — *minimalist RISC-V system emulator capable of running Linux* [GitHub]

- Implemented VirtIO-GPU 2D/3D graphics and keyboard/mouse input via Linux DRM/KMS, VirGL, and an SPSC event queue.

5568ke — *C++20/OpenGL glTF renderer and animation sandbox* [GitHub]

- Built a glTF renderer with scene graphs, keyframe animation, GPU skinning, and ImGui playback controls.

Mase — *maze generation and pathfinding visualizer* [GitHub]

- Built a C++17/OpenGL/ImGui visualizer for three maze generators and five pathfinding algorithms, including UCS and A*.

Talks

Open Source Summit North America 2026 [Talk] [Slides]

- Embedded Linux Conference track — "Demystifying VirtIO-GPU: Building a Graphics Virtualization Bridge From Scratch".

COSCUP 2026 [Talk] [Slides]

- System Software track — "How a VM Draws Pixels: Mesa, Gallium, and virtio-gpu 3D".

Technical Publications

Cpp-Miner [GitHub]

- Authored a Chinese-language series explaining C++ standard terminology and modern C++ features.

Traditional Chinese technical writing

- Translated the xv6 RISC-V book into Traditional Chinese. [GitHub]
- Translated the official Khronos glTF Tutorial into Traditional Chinese. [GitHub]
- Translated *Operating System in 1,000 Lines* into Traditional Chinese. [GitHub]

Skills

Programming: C++, C, Python, Lua
Systems & Graphics: Linux, RISC-V, VirtIO, DRM/KMS, QEMU, OpenGL, Vulkan, Mesa/Gallium, VirGL
Infrastructure & Tools: Docker, Proxmox VE, CMake, Git, GLFW, ImGui
Languages: Mandarin Chinese (native), English (working proficiency), Japanese (basic)